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Studierichting: Informatica
Oefeningenreeks 5G nr. 16.4

$$a = 1 \text{ en } b = 27 \text{ en } r = y^{\frac{1}{3}} - 1$$

$$\begin{aligned} V &= \pi \int_1^{27} (y^{\frac{1}{3}} - 1)^2 dy \\ &= \pi \int_1^{27} (y^{\frac{2}{3}} - 2y^{\frac{1}{3}} + 1) dy \\ &= \pi \left[\frac{3}{5} y^{\frac{5}{3}} - \frac{3}{2} y^{\frac{4}{3}} + y \right]_1^{27} \\ &= \pi \left(\frac{3}{5} 27^{\frac{5}{3}} - \frac{3}{2} \cdot 27^{\frac{4}{3}} + 27 \right) - \pi \left(\frac{3}{5} 1^{\frac{5}{3}} - \frac{3}{2} \cdot 1^{\frac{4}{3}} + 1 \right) \\ &= \pi \left(\frac{729}{5} - \frac{243}{2} + 27 \right) - \pi \left(\frac{3}{5} - \frac{3}{2} + 1 \right) \\ &= \pi \left(\frac{513}{10} - \frac{1}{10} \right) \\ &= \frac{512\pi}{10} \\ &= \frac{256\pi}{5} \end{aligned}$$

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References